

Lab Assignment Task

JavaScript Introduction

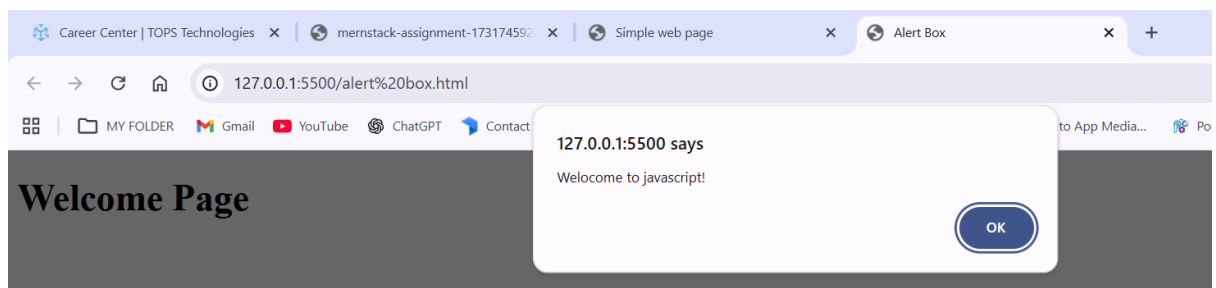
- 1) Create a simple HTML page and add a `<script>` tag within the page.

Ans:-

Welcome to my first javaScripe page

- 2) Write JavaScript code to display an alert box with the message "Welcome to JavaScript!" when the page loads.

Ans:-

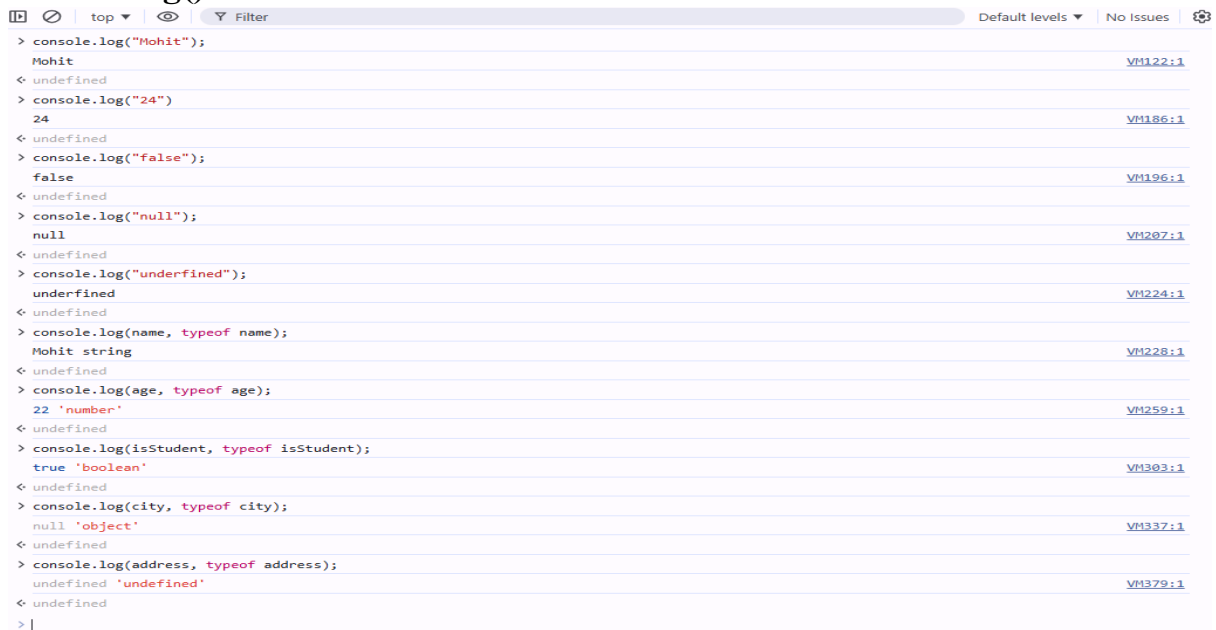


Variables and Data Types

- 1) Write a JavaScript program to declare variables for different data types (string, number, Boolean, null, and undefined).

Ans:- Same 2

2) Log the values of the variables and their types to the console using `console.log()`. Ans:-



```
> console.log("Mohit");
Mohit
VM122:1
< undefined
> console.log("24");
24
VM186:1
< undefined
> console.log("false");
false
VM196:1
< undefined
> console.log("null");
null
VM207:1
< undefined
> console.log("underfined");
underfined
VM224:1
< undefined
> console.log(name, typeof name);
Mohit string
VM228:1
< undefined
> console.log(age, typeof age);
22 'number'
VM259:1
< undefined
> console.log(isStudent, typeof isStudent);
true 'boolean'
VM303:1
< undefined
> console.log(city, typeof city);
null 'object'
VM337:1
< undefined
> console.log(address, typeof address);
undefined 'undefined'
VM379:1
< undefined
> |
```

JavaScript Operators

1) Create a JavaScript program to perform the following:

- Add, subtract, multiply, and divide two numbers using arithmetic operators.
- Use comparison operators to check if two numbers are equal and if one number is greater than the other.
- Use logical operators to check if both conditions (e.g., $a > 10$ and $b < 5$) are true.

Ans:- .

- Arithmetic operators: +, -, *, /
- Comparison operators: === (equal), > (greater than)
- Logical operator: && checks if both conditions are true

Control Flow (If-Else, Switch)

Task 1: • Write a JavaScript program to check if a number is positive, negative, or zero using an if-else statement.

Ans:- It takes input from the user.

❖ Checks:

- $> 0 \rightarrow$ Positive
- $< 0 \rightarrow$ Negative
- Otherwise $\rightarrow$ Zero

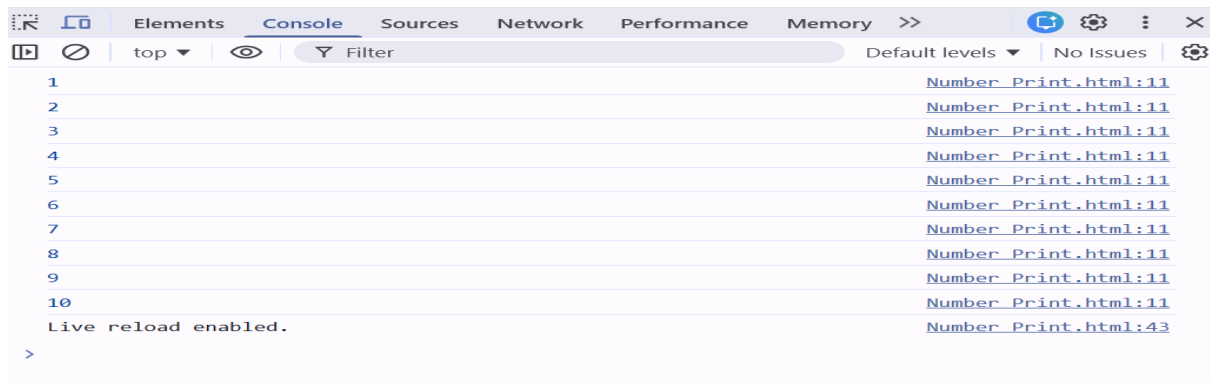
Task 2: • Create a JavaScript program using a switch statement to display the day of the week based on the user input (e.g., 1 for Monday, 2 for Tuesday, etc.).

Ans:- The switch statement matches the number and prints the corresponding day

Lab Assignment

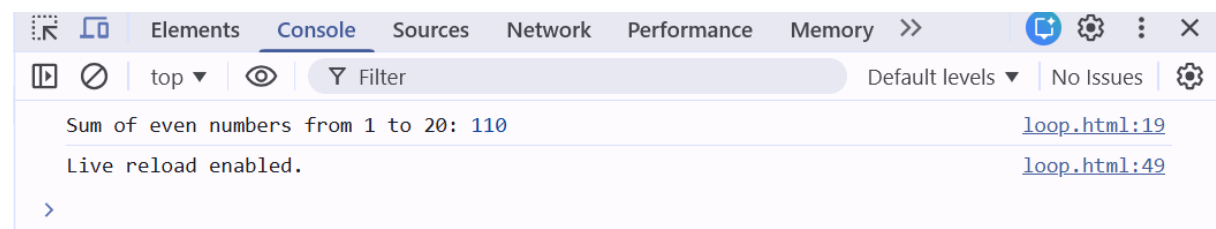
• Task 1: • Write a JavaScript program using a for loop to print numbers from 1 to 10.

Ans:-



• Task 2: • Create a JavaScript program that uses a while loop to sum all even numbers between 1 and 20.

Ans:- This starts from 2 (first even number), keeps adding even numbers, and stops at 20.



• Task 3: • Write a do-while loop that continues to ask the user for input until they enter a number greater than 10.

Ans:- → The do-while loop runs at least once.

→ It keeps prompting the user until the entered number is greater than 10.

You entered: 5-10

Functions

Lab Assignment

- **Task 1:** Write a function greet User that accepts a user's name as a parameter and displays a greeting message (e.g., "Hello, John!").

Ans:-

```
Hello, John!
```

```
>
```

- **Task 2:** Create a JavaScript function calculate Sum that takes two numbers as parameters, adds them, and returns the result.

Ans:-

```
30 calculate.html:14
```

```
Live reload enabled. calculate.html:44
```

```
> |
```

Arrays

Lab Assignment

- **Task 1:** • Declare an array of fruits (["apple", "banana", "cherry"]). Use JavaScript to: • Add a fruit to the end of the array. • Remove the first fruit from the array. • Log the modified array to the console.

Ans:-

```
▼ Array(3) array.html:21  
  0: "banana"  
  1: "cherry"  
  2: "orange"  
  length: 3  
  ► [[Prototype]]: Array(0)
```

- **Task 2:** • Write a program to find the sum of all elements in an array of numbers.

Ans:-

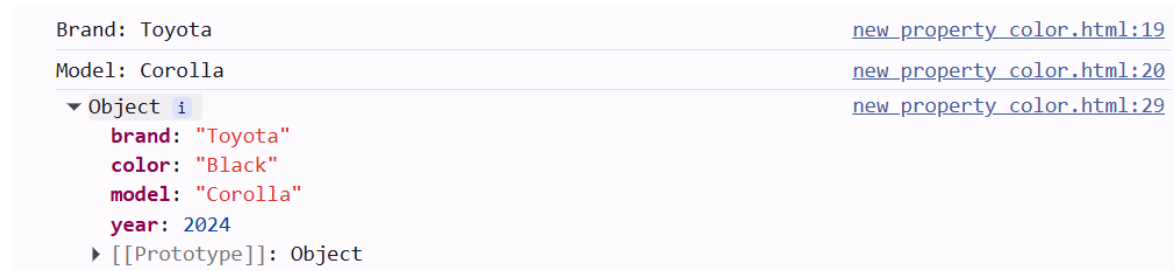
```
Sum of array elements: 150 sum.html:21
```

Objects

Lab Assignment

• Task: • Create a JavaScript object car with properties brand, model, and year. Use JavaScript to: • Access and print the car's brand and model. • Update the year property. • Add a new property color to the car object.

Ans:-



JavaScript Events

Lab Assignment

• Task: • Create a simple webpage with a button that, when clicked, displays an alert saying "Button clicked!" using JavaScript event listeners.

Ans:-



DOM Manipulation

Lab Assignment

• Task: • Create an HTML page with a paragraph () that displays "Hello, World!". • Use JavaScript to: • Change the text inside the paragraph to "JavaScript is fun!". • Change the color of the paragraph to blue.

Ans:-

JavaScript is fun!

JavaScript Timing Events (setTimeout, setInterval)

Lab Assignment

- Task 1: • Write a program that changes the background color of a webpage after 5 seconds using setTimeout().

Ans:-

Background color will change after 5 seconds

- Task 2: • Create a digital clock that updates every second using setInterval()

Ans:-

Digital Clock

10:53:51

JavaScript Error Handling

Lab Assignment

- Task: • Write a JavaScript program that attempts to divide a number by zero. Use trycatch to handle the error and display an appropriate error message.

Ans:-

Division by Zero Example

Error: Cannot divide by zero.